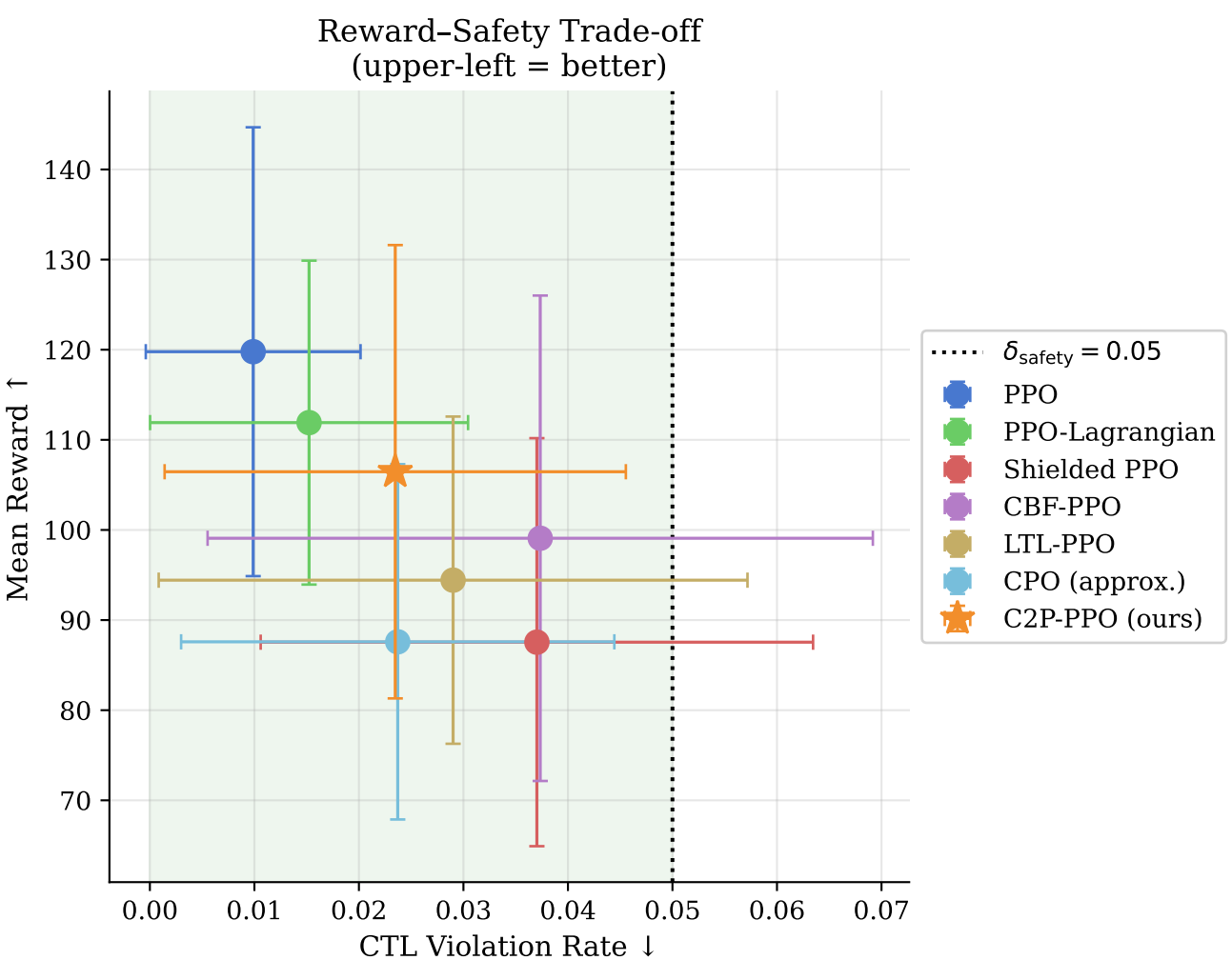
Reward–Safety Trade-off
(upper-left = better)
Mean Reward ↑
CTL Violation Rate ↓
140
130
120
110
100
90
80
70
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
$\delta_{\text{safety}} = 0.05$
PPO
PPO-Lagrangian
Shielded PPO
CBF-PPO
LTL-PPO
CPO (approx.)
C2P-PPO (ours)